Algorithm 1: Grade submissions with a local LLM

Input : directory, filename_pattern, rubric_prompt, model, force
Output: one JSON + one TXT per submission, plus summary.csv

$results \leftarrow \emptyset$;
foreach *submission* $\in$ `find_submissions(`*directory, filename_pattern*`)` **do**
    if `already_graded(`*submission*`)` **and not** *force* **then**
        continue ; `// cache hit --- no Ollama call made`
    $(text,\ wc,\ pc,\ ic) \leftarrow$ `extract(`*submission.path*`)`; `// local only: PyMuPDF / python-docx / openpyxl`
    prompt $\leftarrow$ `build_user_prompt` (*facts* $= (wc, pc, ic)$, *text* $=$ *text*) ;
    reply $\leftarrow$ `call_ollama` (*system* $=$ *rubric_prompt, user* $=$ *prompt, model* $=$ *model,*
        url $=$ `http://localhost:11434/api/chat`);
    `// the one network call, loopback only`
    result $\leftarrow$ `parse_json(`*reply*`)`;
    `save(`*submission, result*`)`;
    $results \leftarrow results \cup \{result\}$;
`write_summary_csv(`*results*`)`;